



University of Essex

Online

Research Proposal Presentation Transcript

**Topic: Cybersecurity threats in IoT for the Power and Utilities
sector Smart Grids and SCADA**

Module: Research Methods and Professional Practice

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Slide 1 – Title

Hello. My name is Abdulrahman Husain Ahmed Alhashmi. This talk presents my research proposal. It examines cybersecurity threats from IoT in power and utilities. The focus falls on smart grids and SCADA.

The main problem stands out clearly. Utilities add IoT devices and cloud-linked services. They seek better visibility and control. These changes grow the attack surface. They open new failure points too. Such risks hit availability, integrity, and safety.

I cover the research problem. Questions and objectives come next. Key literature and standards follow. Then research design, ethics, the artefact, and timeline.

Slide 2 – Research problem and significance

Utilities digitise field devices, substations, control centres, and enterprise services. They link operational technology to IT networks and third-party services. IoT devices, gateways, and remote paths join the control setup. This includes sites hard to reach.

The problem lies not in missing high-level advice. Strong standards exist for OT and smart grid security. Think NIST smart grid rules and IEC standards. The real gap sits in operations. Utilities lack solid, real-world guidance. It must tie common IoT threats to fixes. These fixes work without halting key services. Patching slots stay short. Legacy gear limits crypto and changes.

The work holds practical and academic value. It aids safe operations. It cuts compromise odds and caps incident damage. In research, it yields tested proof on control choices under real limits. It skips broad advice.

Slide 3 – Aim and objectives

My aim: create and test a practical security check and fix method. It suits IoT smart grids and SCADA under operation limits, The objectives build in steps.

First, map assets and trust lines in a sample setup.

Second, spot and rank threat cases on integrity, availability, privacy.

Third, craft layered defences from known rules. Zoning leads. Strong ID and access control. Safe update routes. Protocol-based checks.

Fourth, set up a small testbed. Build an artefact pack for reuse.

Last, test with attack simulations and ops measures. Refine what succeeds or fails.

Slide 4 – Research questions

Four questions steer the work.

RQ1: What threat cases repeat in IoT smart grids and SCADA? What lets them happen?

RQ2: Which controls cut risk best with OT limits? Like uptime needs and short fix windows.

RQ3: How does ID and key handling grow for AMI? Without hurting trust.

RQ4: What proof lacks in current work? How does testbed testing build it?

These keep focus on real gains and clear results.

Slide 5 – Background architecture and trust boundaries

The study picks a real smart grid and SCADA setup. It marks trust lines clear.

Field layer holds sensors, meters, IEDs, RTUs. They touch real processes. Substation layer has gateways, relays, station nets. They link to ops nets. Control centre runs SCADA servers, historians, EMS, DMS. They aid watch and send orders. Enterprise and cloud handle analytics, bills, vendor aid.

Controls go at lines. Field to substation nets. Substation to control centres. OT to enterprise or cloud. This view sets the threat model and fix picks.

Slide 6 – Threat landscape and scenarios

Threats sort by layer. This fits utility risk handling.

Field or edge sees weak logins, shared accounts, unsafe firmware, body access risks. Time tricks hit power logic and event links.

Substation and net layers have flat nets for side moves. Old protocols let replays or command slips sans checks. Weak remote paths turn pivot spots.

AMI and back-end face key scale and onboard woes. It tests security and uptime. Meter data leaks habits. Supply updates and cloud splits raise risks.

These stem from real OT advice and smart grid studies.

Slide 7 – Key literature and standards, and why the gap remains

Literature agrees on solid practice.

Defence in depth rules. Zoning and set flows. IEC 62351 protocol rules. Signed firmware paths. Strict change rules. OT-fit checks and remote handling.

Yet proof stays thin. Few long data sets match substations. AMI keys prove tough. Old gear blocks crypto, patches. Studies skip side-by-side control tests under load and limits.

Field shifts show this. Early work hit protocol flaws, layouts, net links. New work eyes IoT gates, cloud math, supply risks, remote growth. This study fits there. It makes real evidence, not just ideas.

Slide 8 – Methodology and research design

Design Science Research fits. It builds and tests an artefact.

Phase one: needs and threat model. Turn standards, limits, lit into firm needs. Link threats to setup lines.

Phase two: artefact design. A threat-to-fix kit. Checklists, zoning plans, check rules. Maps threats to do-able fixes.

Phase three: testbed setup. Makes sample layout with SCADA-like traffic, logs. No live ties.

Phase four: tests. Attack plays for logins, replays, command hits, update tricks, time fakes. Check base vs layers.

Phase five: review, tweak. Weigh security and perf. Add user input if possible. Fix artefact from finds.

This yields a useful tool and solid research proof.

Slide 9 – Data collection and analysis plan

Data collection links to the evaluation goals.

From the testbed, I gather network traces for IEC-104 or DNP3 traffic. I collect host and app logs from mock SCADA parts. I also take alerts from monitoring rules.

If practitioner interviews occur, they stay brief. They target feasibility and real-world fit. For example, can controls fit under standard utility change limits?

For analysis, descriptive stats cover test metrics. Clear charts show before-and-after views. If interviews provide data, thematic coding sorts of responses. This handles feedback in a structured way, not as loose stories.

Slide 10 – Artefact(s) to be created

The key artefacts are a Smart Grid and SCADA IoT Security Toolkit.

It holds a zones and flow template. It has a hardening checklist based on NIST and IEC advice. It includes protocol rules for monitoring. A remote access policy template sits there too. Plus, a testbed plan with a small data set for repeat tests.

The toolkit serves two roles. First, it aids teams in assessing or planning IoT links in OT setups. Second, it acts as a research tool for copies and side-by-side tests.

This focus turns the work from mere description to building and testing.

Slide 11 – Evaluation, thresholds, and acceptable operational impact

Evaluation splits into verification and validation.

Verification checks if the artefact runs as planned. It tests configs, scripts, and rule logic. It confirms the testbed shows expected actions.

Validation tests real-scenario help. I compare base states to added controls with attack sims. I track detection reach, detection time, and false alerts.

To match real ops, I measure impacts like latency adds, resource use, and issues such as missed polls or shaky links from monitoring or splits.

I set thresholds in three steps.

First, I log base levels in the testbed for poll success, latency, and normal alerts.

Second, OT guides and sector habits set time and load limits.

Third, practitioner views shape "acceptable." They check if controls deploy without outages or heavy operator load.

In practice, acceptable means no drop in poll success from base. No link issues. Alert rates stay easy for daily work.

Slide 12 – Ethics considerations and risk assessment

Ethics and safety matter most in critical infrastructure security work.

Tests stay in a closed lab. No touch of live utility nets or real gear. Data comes synthetic or stripped of names, key for AML traces.

For interviews, I seek consent. Questions stick to feasibility. No names kept. Data stored safe with tight access.

Risk checks cover study limits too. Testbeds lack full realness, so I note limits and stress comparisons over broad claims. Scope growth threatens, but questions and metrics hold the line.

Slide 13 – Timeline

The plan spans twelve weeks.

First weeks sharpen questions, file ethics forms, and finish threat models and needs.

Mid weeks build the testbed, craft the first artefact, and run tests.

Late weeks handle analysis, charts, result writes, and prep for talks and submits.

Milestones mark ethics okay, working testbed, full artefact, and tested results with tips.

Slide 14 – Expected contributions and close

Contributions come in three parts.

First, a threat-to-control map tied to true utility limits, not vague ideas.

Second, solid proof of security-ops trade-offs from testbed runs.

Third, a hands-on toolkit for academic checks and practitioner tweaks. It lists evidence gaps to steer next steps, like data sets and AMI key handling.

Thank you for your time. Feedback welcome on scope, eval setup, and toolkit value for ops.

Slide 15– References

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Thank you